

from this, centralisation of power supply usually leads to a higher voltage drop at connection circuits.

That is a result of supplying larger power at the input and further remoteness of consumers at the output, which makes it necessary to use wires of larger cross-section. A decentralised power supply system has lower powers transferred inside, while the power supply units can be located close to their consumers. Thus it minimises dynamic instabilities of the voltage.

Another advantage of power supply units for decentralised systems is their unified character, i.e., these can be used in power supply systems of different devices, unlike high-power systems, which are usually unique and can be used for one type of equipment only.

If a decentralised system uses PSUs with output current higher than pulse current of the supplied load, the specific drop of output voltage in the end of the operating pulse will be fully eliminated in this case.

First, this feature eliminates the need to use large capacitance accumulators for power supply to the load during the complete power consumption pulse, retaining the minimum capacitance to ensure the required voltage quality during the transition process in the front and rise of load pulse.

Second, as the mode of nominal power of the converter corresponds to its maximum efficiency, thermal losses of the complete power supply system are minimised.

Third, in case of AESA transceivers, which usually have pulse off-duty factor of less than 5, the comfortable thermal mode of the unit itself is supported.

An important advantage of a PSU with fast voltage feedback is galvanically isolated differential synchronisation input. This allows frequency synchronisation of several simultaneously operating units and ensures reliable hardware and software filtration of electromagnetic interference of converters. High conversion frequency not only positively impacts the weight and dimensions of the units but also allows to isolate in different ranges the operating frequency area of the AESA transceiver equipment and units' switching noise.

When selecting a power supply unit with fast voltage feedback, it is important to consider small dimensions as well as the ability to operate within as wide a temperature range as possible, which sometimes can reach -60 to +125°. Other important functions may include remote on/off, protection against over-current, short-circuit and over-voltage, as well as output voltage fault monitoring. Polymer potting sealing ensures strong protection of the units against damage from harsh environments, vibration, dirt, moisture or salt mist. **EFY**



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